

Provenance notice. This report distinguishes two classes of number, and never mixes them in a single claim. **MEASURED** = produced by SabiLine's own evaluation harness against live model APIs during this submission. **PUBLISHED** = figures from the supplied Intron AfriHealth MultiBench reference documentation, cited for comparison and *not* re-derived here. Sahara's ASR endpoint (`api.intron.io`) was unreachable from the sandboxed build network (HTTP 403 at the egress proxy), so Sahara appears here with published figures only; §6 gives the one-command path to reproduce it on an unrestricted network.

1. Model comparison summary

Model	Macro WER	Macro CER	Latency	Afr. coverage	Pros / Cons
Intron Sahara PUBLISHED	0.244	0.107	412 ms	96.4%	+ Best tonal-language resilience; trained natively on W/E African clinical audio; widest language coverage. – Vendor-specific API; could not be independently verified in this environment.
Meta OmniLLM PUBLISHED	0.268	0.119	580 ms	88.5%	+ Strong multimodal speech reasoning on accented multilingual speech. – Moderate latency at code-switch boundaries; 2nd-tier tonal accuracy.
Azure Cognitive Speech PUBLISHED	0.281	0.124	340 ms	62.0%	+ Fastest; best-in-class on standard Nigerian English (0.168). – Coverage collapses on tonal languages (Igbo 0.368); weakest African language breadth.
Google Gemini PUBLISHED (figures at left)	0.295	0.131	510 ms	84.0%	+ Only model here that does language ID + transcription + dialogue in one call; measured medical-term retention perfect (0 drug-name errors, §5a). – Sensitive to tonal orthography; free-tier rate limits throttle batch evaluation. <i>This is the model SabiLine runs in production; it is also the only row with independently measured results — see §4a.</i>
Whisper Large-v3 PUBLISHED	0.389	0.182	720 ms	54.5%	+ Open weights; self-hostable at zero marginal cost. – Worst on every African language (Igbo 0.510); hallucination spikes at code-switch junctures; slowest.

Headline: the ranking is driven almost entirely by *African-language* performance, not overall capability. Azure and Whisper are competitive-to-excellent on Nigerian English (0.168 / 0.224) yet fall to 0.368 and 0.510 on Igbo — a 2.2× and 2.3× degradation. Sahara's advantage is not that it is a better ASR system in general, but that its training distribution includes the speech our users actually produce.

2. Data used

Live evaluation corpus **MEASURED**

Source: 12 de-identified clinical instances authored to mirror the AfriHealth MultiBench structure and clinical domains (`src/services/benchmark/sampleDataset.ts`).
Languages (6): Hausa, Yoruba, Igbo, Nigerian English, West-African French, + 4 code-switched instances (EN ↔ HA, EN ↔ YO, EN ↔ IG, FR ↔ EN).
Volume: 111.5 s of audio across 12 instances (8.4–10.4 s each); mean 9.3 s.
Domains: maternal health, malaria/fever triage, cardiology, pharmacy adherence, vaccination.
Privacy: zero PII; de-identified speaker IDs (`SPK_NG_HA_101`); consent-verified and PII-scrubbed flags carried per record.

Reference corpus **PUBLISHED**

Source: Intron AfriHealth MultiBench (supplied competition reference dataset).
Volume: 5,200 instances / ~20 hours / 600 speakers / 19 languages.
Breakdown: 3,200 transcription · 1,600 translation · 398 spoken-QA recordings.
Primary African languages: Hausa, Igbo, Yoruba, Nigerian English, Swahili, Twi, Zulu.
De-identification: PII scrubbing with double-pass clinician + human annotator verification.
Ethics: IRB/consented protocol for research use of de-identified healthcare audio.

Preprocessing & audio protocol

Reference audio is **synthesized once per instance** from the de-identified reference transcript and the **identical waveform is served to every model** under comparison, so no model gets a cleaner signal than another. Text normalization before scoring: punctuation stripped, lower-cased, whitespace collapsed, and **diacritics deliberately preserved** (`stripDiacritics: false`) — stripping them would silently erase the tonal distinctions this benchmark exists to measure. The live intake path additionally applies gain normalization to quiet microphone input before transcription.

Methodological limitation, stated plainly. A synthesized-audio protocol perfectly controls speaker, channel and noise variation — every model is scored on exactly the same signal — but it does *not* reproduce real clinic acoustics: background noise, disfluency, overlapping speech, or genuine speaker accent variation. Absolute WER figures here should therefore be read as **a lower bound** on real-world error, and the protocol's value is in *relative* model comparison, not in predicting field accuracy.

3. Evaluation metrics & why they are appropriate

Metric	Definition	Why appropriate for this task
WER	Word-level Levenshtein edit distance ÷ reference word count.	Industry-standard, comparable to all published ASR literature. But it treats a clitic-merge (jinin ki → jininki) as a full error, which systematically over-penalizes agglutinative and tone-marked African orthographies. Reported, but never alone.
CER	Character-level edit distance ÷ reference character count.	The load-bearing metric for this benchmark. Robust to tokenization and orthographic variance, so it separates "spelled the tone mark differently" from "misheard the drug name" — a distinction WER cannot make in Hausa, Yoruba or Igbo.
Accuracy	1 – WER, floored at 0.	Not a research metric; included because clinic administrators and procurement staff — the people who decide whether to deploy this — read "78% accurate" and not "0.222 WER".
BLEU / chrF	n-gram / character-n-gram overlap against reference translation.	Downstream translation quality. chrF is weighted more heavily than BLEU here because it is markedly more reliable for morphologically rich languages where valid translations differ in surface word form.
AfriCOMET	Learned neural translation metric tuned on African languages.	Correlates with human judgement where BLEU does not, precisely for the low-resource languages in scope.
Clinical error weighting	Counts of medical-term errors, named-entity errors, and code-switch boundary errors.	The metric that actually matters for patient safety. A 20% WER spread across function words is an annoyance; a 20% WER concentrated in drug names, dosages or allergies is a clinical incident. These are counted separately so the two can never be confused.

4. Quantitative results

4a. Word / Character Error Rate per language

Model	Hausa	Igbo	Yoruba	Nig. English	Afr. French	Macro CER
Intron Sahara PUB	0.218	0.252	0.231	0.185	0.260	0.107
Meta OmniLLM PUB	0.249	0.284	0.262	0.198	0.272	0.119
Azure Cognitive PUB	0.334	0.368	0.312	0.168	0.245	0.124
Google Gemini PUB	0.289	0.315	0.298	0.174	0.252	0.131
Whisper Large-v3 PUB	0.442	0.510	0.428	0.224	0.298	0.182
Gemini 3.8 Flash MEASURED SabiLine harness, live API, n=1 instance	WER 0.222 CER 0.025	not measured — free-tier rate limit (≈1 instance/min, 20 req/day) each instance costs 2 API calls (synthesis + transcription)				—

Reading the measured row: Gemini scored WER 0.222 but CER 0.025 on the same utterance — a **9× gap**. That gap is the single most important quantitative finding in this report, and §5 explains what produced it. Latency 2,677 ms end-to-end (includes cold start); 18 reference words, 81 reference characters; 3 substitutions, 1 deletion, 0 insertions; **0 medical-term errors, 0 named-entity errors, 0 code-switch boundary errors**.

4b. Downstream task — clinical translation quality PUBLISHED

This is the task the code-switched transcript feeds: a transcript is only useful here if it can be turned into instructions a patient can act on, in their own language.

Direction	BLEU	chrF	AfriCOMET	Clinical domain
English → Hausa	0.284	0.532	0.684	Clinical triage & patient instructions
English → Yoruba	0.261	0.518	0.662	Antenatal & community health education
English → Igbo	0.253	0.498	0.648	Chronic disease & hypertension guidance
English → French	0.385	0.642	0.792	Public health announcements

The ~13-point BLEU and ~14-point AfriCOMET gap between French and the three Nigerian languages is the resource gap made numeric: the same pipeline, same domain, same clinical register, ~50% better output for the language with orders of magnitude more training data. Igbo is consistently last across all three metrics.

5. Qualitative findings

5a. The measured case, in full MEASURED

REFERENCE (ha) — AHMB-HA-001, antenatal iron/folate counselling

Kina bukatar shan maganin iron da folic acid a kowace rana don karfafa jinin ki da lafiyar jaririn.

GEMINI HYPOTHESIS — WER 0.222 / CER 0.025

kina bukatar shan maganin iron da folic acid a kowace rana don karfafa jinkin da lafiyar jaririn

Every single counted error is orthographic, not acoustic:

- **k vs k (2 errors).** The model wrote the hooked consonant k in bukatar and karfafa. This is **correct standard Hausa orthography** — the reference used the ASCII-fallback k. The model was penalized for being *more* orthographically correct than its ground truth.
- **Clitic merge (1 substitution + 1 deletion).** jinin ki ("your blood") written as the equally valid bound form jinki. Two word-level errors for a whitespace decision that changes nothing a patient or clinician would hear or understand.
- **Clinical payload: perfect.** iron, folic acid, kowace rana (daily frequency) — the drug names, the dosage schedule, the entire actionable content — transcribed without error.

Implication for deployment: had we ranked models on WER alone, we would have recorded a 22% error rate on an utterance that is clinically flawless. Any procurement decision or model-selection process for African-language clinical ASR that reports WER without CER alongside it is measuring orthographic convention and calling it accuracy.

5b. Per-model, per-language characteristics PUBLISHED

Model	Strengths	Weaknesses	Language-specific notes
Intron Sahara	Most consistent across all five African languages; narrowest spread between best and worst language (0.185 – 0.260, a 1.4× range).	Not independently verifiable here; closed API; single-vendor dependency.	HA best-in-class 0.218. IG 0.252 — the only model to hold Igbo under 0.26. YO 0.231. Tonal resilience is the differentiator.
Meta OmniLLM	Solid second across the board; strong on accented multilingual speech.	Code-switch boundary latency; no language leads outright.	IG 0.284 is its weakest. Consistent but never first.
Azure Cognitive	Fastest (340 ms); best Nigerian English of any model (0.168); best African French (0.245).	Coverage 62% — the narrowest. Degrades hardest on tonal languages.	EN excellent. IG 0.368 and HA 0.334 — a 2.2× penalty vs its own English. Viable only for English-dominant deployments.
Google Gemini	Only model providing language-ID + transcription + conversational reply in one pipeline — which is what removes the language picker from SabiLine entirely. Medical terminology retention measured perfect.	Tonal/diacritic orthographic sensitivity; free-tier rate limits make batch evaluation impractical; single point of failure (its TTS also synthesizes the benchmark audio).	HA measured CER 0.025 — near-perfect at character level. IG/YO published 0.315/0.298: tonal orthography remains its weak axis.
Whisper Large-v3	Open weights; self-hostable; no per-call cost; full data control for privacy-sensitive deployments.	Worst on every African language; hallucinates at code-switch junctures — the most dangerous failure mode here, since it produces fluent, confident, wrong text rather than an obvious error.	IG 0.510 — one word in two. HA 0.442, YO 0.428. Not deployable for Nigerian-language clinical intake without fine-tuning.

6. Limitations & reproduction

Known limitations

- **n=1 measured instance.** Free-tier quota (~1 instance/min, 20 requests/day; 2 calls per instance) capped live evaluation. The measured figure is a verified single observation, not a distribution — no variance or confidence interval is claimed.
- **Sahara unverified locally** — egress-blocked (HTTP 403), not a code defect; the integration is implemented and configured.
- **Synthesized, not recorded, audio** — see §2. Absolute figures are a lower bound on field error.
- **Model B / Model C** ship as an honest bring-your-own-endpoint extension point and report "not configured" rather than emitting placeholder numbers.

Reproducing this run

Set `GEMINI_API_KEY` and `SAHARA_API_KEY`, then from an unrestricted network:

```
curl -X POST $HOST/api/benchmark/run \
-H "Content-Type: application/json" \
-d '{"sampleIds":["AHMB-HA-001", ...],
  "selectedModels":["gemini","sahara"]}'
```

Or use the in-app **Benchmark** tab (?studio=1), which exposes per-instance selection, model selection, and the normalization switches. Every metric in §4a is computed by `src/services/benchmark/{wer,cer,accuracy}.ts` using standard Levenshtein edit distance with no weighting or score adjustment; the code is open in the repository.